EX. No.3.1. CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

DATE:

AIM:

To create a 3D model from given orthographic projection

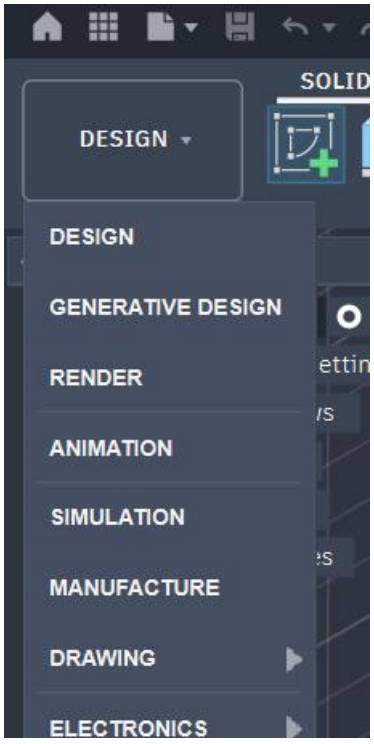
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REQUIREMENTS:

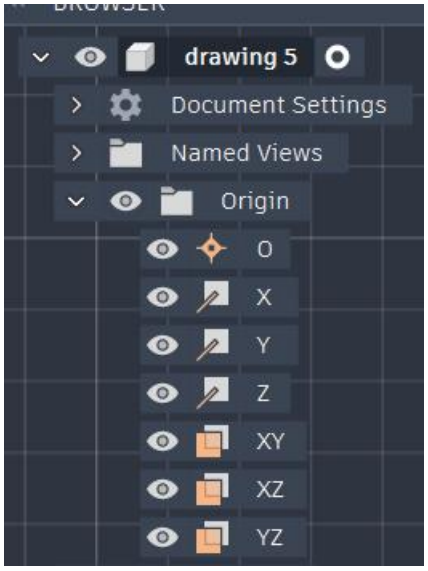
1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

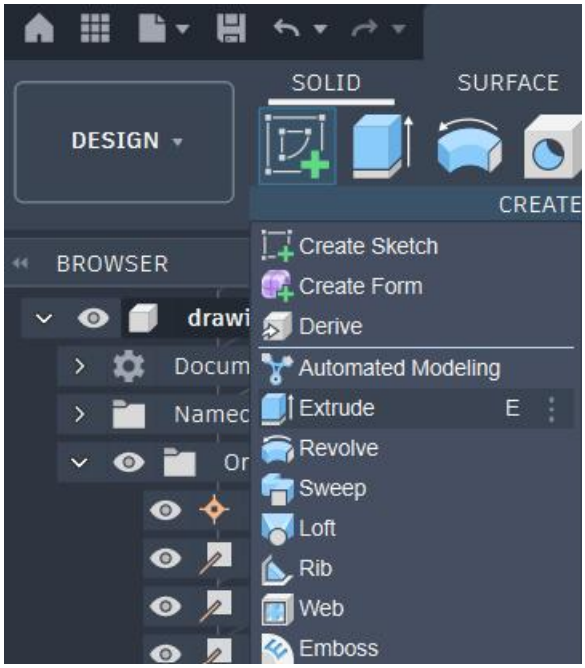
1. First, open Auto desk Fusion 360 and create a new design.



2. Select the XY plane and start a 2D sketch.

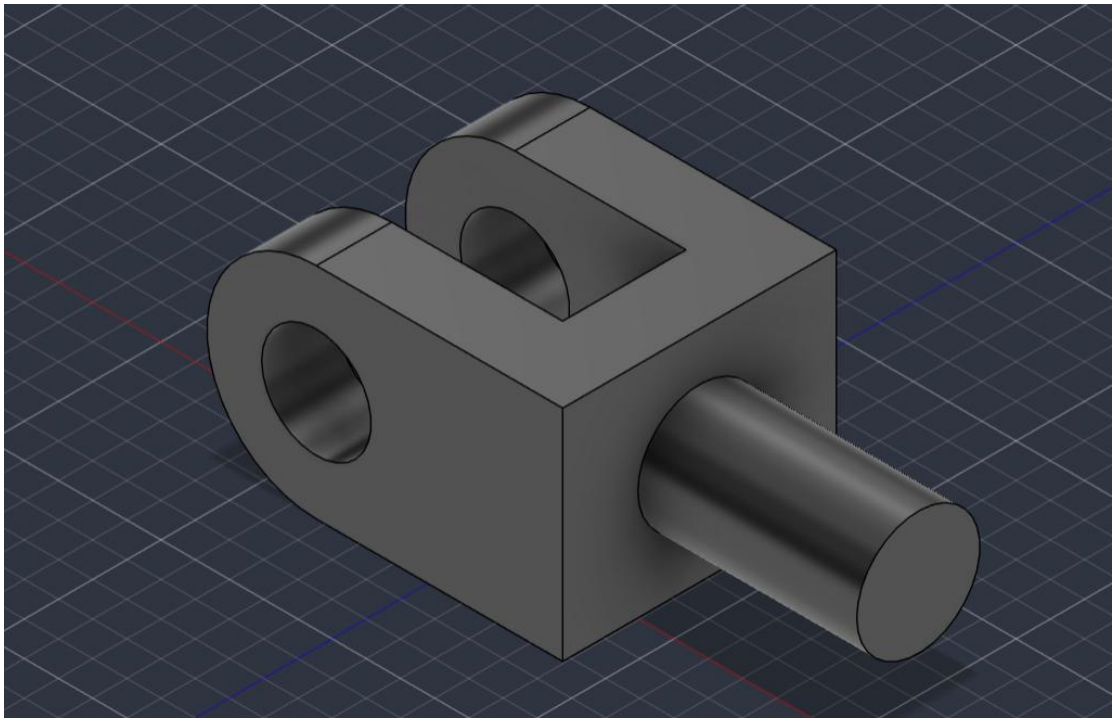


3. Draw the main body with a width of 40 mm and a length of 36 mm from the centre.
4. Draw the front rounded portion using an arc of radius 16 mm (R16).
5. Draw a circular hole at the centre as shown in the drawing
6. Create the fork-shaped portion with two arms of 11 mm thickness each and a gap of 18 mm between them.
7. Set the overall width as 40 mm and the vertical dimension as 32 mm.
8. Draw the extended rectangular portion with a length of 32 mm.
9. Apply the remaining dimensions of 52 mm and the given hole dimensions (Ø16 and Ø18) as shown in the drawing.
10. Use the Extrude and Cut commands wherever required to complete the 3D model.

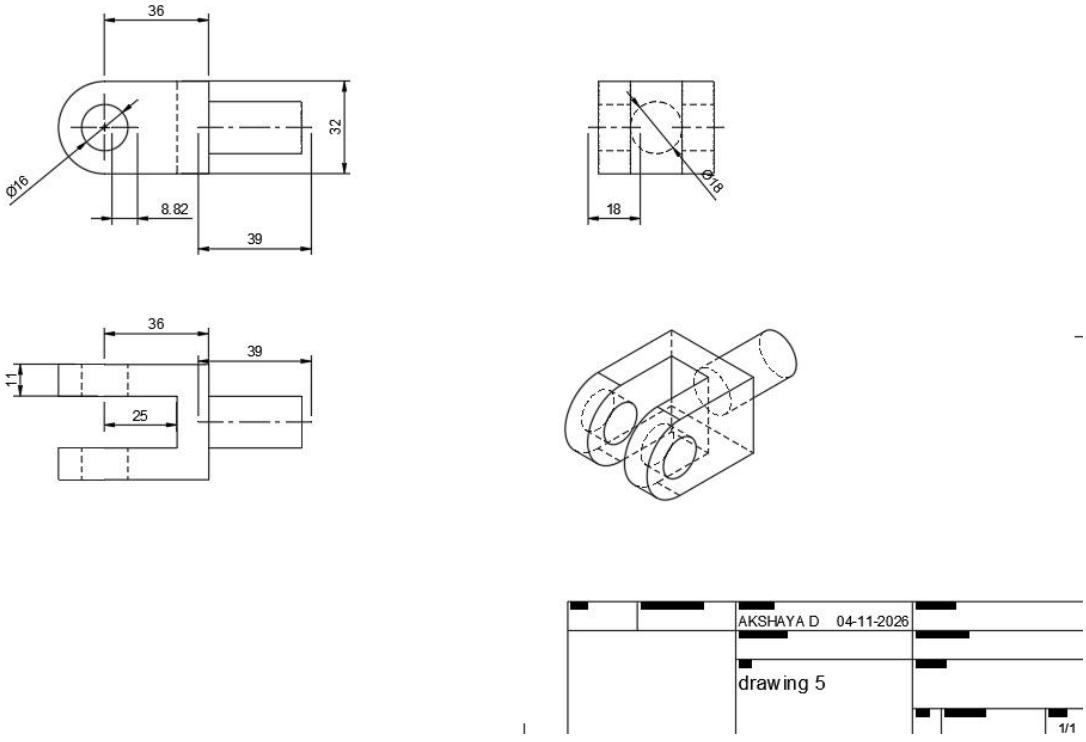


11. Finally, check the model with the given drawing and save the design.

INPUT:



OUTPUT:



RESULT:

Thus, 3D model is created from orthographic projection.